

SEC P vs NP log 2026-05-18

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 10:27 UTC

SEC P vs NP — daily log 2026-05-18

16 cycles recorded

Signed-Laplacian Holonomy Spectrum Bounds Tseitin Tree-Resolution

- **Verdict:** INCONCLUSIVE
- **Bridge:** Discrete magnetic / signed Laplacian spectral theory of finite graphs — the $\mathbb{Z}/2$ Bochner-Laplacian L^σ for a line bundle on G representing a cohomology class in $H^1(G; \mathbb{F}_2)$, accessed through the gauge-invariant smallest eigenvalue $\lambda_{\min}(L^\sigma)$ (Lieb-Loss 1993 on magnetic operators; Sunada 1994 on discrete magnetic Schrödinger; Mathai-Yates 2006; Atay-Liu 2014 and Lange-Liu-Peyerimhoff-Post 2015 on signed Cheeger inequalities). An arXiv search for ‘magnetic Laplacian’ AND (‘Tseitin’ OR ‘resolution refutation’ OR ‘proof complexity’) returns 0 direct hits and <5 adjacent papers, none using gauge-invariant signed spectra as a CNF invariant; distinct from sandpile / 2-Sylow / Ihara-zeta attempts on Tseitin (those use cokernels, non-backtracking matrices, or critical-group orders, never the $\mathbb{F}_2$ Bochner spectrum). $\times$ Tree-like Resolution / DPLL refutation size $t(T(G, \omega))$ for Tseitin XOR formulas $T(G, \omega)$ on connected 3-regular graphs G with odd charge ω (Urquhart 1987 / Ben-Sasson-Wigderson 2001 / Alekhnovich-Razborov 2001 expansion regime), in the Cook-Reckhow-Krajíček bounded-arithmetic framework where $\neg T(G, \omega)$ is Σ^b_0 and $\log_2 t$ controls V^0 -witnessing complexity.
- **Recorded:** 2026-05-18 00:33 UTC
- **Entry ID:** 2d264de23a0e

Statement

For a connected 3-regular graph $G=(V,E)$ with odd charge $\omega:V\rightarrow\mathbb{F}_2$ ($\sum_v \omega(v)\equiv 1 \pmod 2$), build a $\mathbb{Z}/2$ -gauge representative $\psi\in\mathbb{F}_2^E$ by BFS from a fixed root v_0 , setting each tree edge so that $\oplus_{e\in v}\psi(e)=\omega(v)$ at every non-root v (the unsatisfiability deficit lands at v_0); form the signed adjacency A^σ with $\sigma(e)=(-1)^{\psi(e)}$ and the signed Laplacian $L^{\sigma=D-A^\sigma}$, and define $\mu(G,\omega):=\lambda_{\min}(L^\sigma)$, which is gauge-invariant under $\psi\mapsto\psi\oplus\delta f$ and equals 0 iff $\sum\omega$ is even. We conjecture an absolute constant $c>0$ such that for every unsatisfiable $T(G,\omega)$ on 3-regular G , $\log_2 t(T(G,\omega))\geq c\cdot\mu(G,\omega)\cdot|V|$, with pre-registered $c=0.05$. A single 3-regular instance (G,ω) with $\log_2 t(T(G,\omega))<0.05\cdot\mu(G,\omega)\cdot|V|$ falsifies the conjecture.

Rationale

The Tseitin obstruction is exactly the $\mathbb{Z}/2$ holonomy class $[\omega]\in H^1(G;\mathbb{F}_2)$, and $\mu(G,\omega)$ is the spectral norm of this obstruction in the discrete Bochner-Laplacian sense — large ($\Theta(1)$) on expanders where holonomy is globally rigid, and small ($\Theta(1/|V|^2)$) on tree-like graphs where the obstruction concentrates locally. The mechanism funnels the Ben-Sasson-Wigderson expansion-based $\exp(\Omega(\text{boundary expansion}))$ lower bound through a single gauge-invariant scalar via the signed Cheeger inequality $h^\sigma(G)\geq\sqrt{(2d\mu)}$, making the standard expansion bound a corollary of a finer structural invariant.

The invariant is a property of the CNF instance (not of any truth-table), so it is shielded from natural proofs; it does not algebrize (signed Laplacian eigenvalues are not preserved under polynomial extension of the Boolean ring); and Tseitin lower bounds for tree-Resolution are not relativizing-style oracle facts.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.12s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92felba4.py", line 158, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92felba4.py", line 126, in run_trial
    psi = bfs_gauge(vertices, edges, 0)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92felba4.py", line 111, in bfs_gauge
    if adj[u][v] == 1 and not visited[v]:
       ^^^
```

NameError: name 'adj' is not defined

Judge reasoning

Test crashed due to undefined variable 'adj' in BFS gauge calculation, preventing evaluation of conjecture validity | next: Fix the 'adj' variable definition in bfs_gauge function and rerun tests with multiple seeds to assess support fraction

Bonami-Beckner 4-2 Saturation of Clause-Sum Bounds DPLL Depth

- **Verdict:** INCONCLUSIVE
- **Bridge:** Bonami-Beckner hypercontractivity (Bonami 1970, Beckner 1975) — the degree-d 4-norm-vs-2-norm contraction $\|P\|_4 \leq (q-1)^{\{d/2\}} \|P\|_2$ — applied as a SYNTACTIC Fourier-analytic invariant of the centered clause-indicator polynomial Q_F of a CNF (a property of the formula's clause/polarity structure, NOT of its truth table). An arXiv search 'Bonami-Beckner hypercontractivity' AND ('DPLL' OR 'tree-Resolution' OR 'CNF refutation') returns 0 direct hits, with <5 adjacent papers (Schoenebeck-style PC-degree, never DPLL search-tree depth via syntactic clause-sum 4-2 ratios). Distinct from the blacklisted Cross-Side Fourier attempt: that invariant was computed on the truth-table communication MATRIX (natural-proof shape); here Q_F lives on the clause-polarity STRUCTURE of F , so two semantically equivalent CNFs F_1, F_2 can have very different $r(F)$, shielding the invariant from Razborov-Rudich constructivity. $\times$ DPLL / tree-like Resolution refutation depth $d(F)$ for unsatisfiable 3-CNFs, in the Ben-Sasson-Wigderson-Krajíček bounded-arithmetic regime where $\log_2 d(F)$ controls V^0 -witnessing complexity of $\neg F$ as Σ^b_0 and lower-bounds tree-Resolution width $w^*(F)$.
- **Recorded:** 2026-05-18 03:29 UTC
- **Entry ID:** 2b2653395609

Statement

For an unsatisfiable 3-CNF F with m clauses on n variables, let $s_{\{i,v\}} \in \{\pm 1\}$ be the polarity sign of variable v in clause C_i ($s=+1$ if v occurs negated, -1 if positive). Define the centered

clause-falsification polynomial $Q_F: \{-1, +1\}^n \rightarrow \mathbb{R}$ by $Q_F(x) = \sum_{i=1}^m (\prod_{v \in \text{vars}(C_i)} (1 + s_{i,v} x_v) / 2 - 1/8)$, whose Fourier coefficient at $T \subseteq [n]$ with $|T| \in \{1, 2, 3\}$ is the closed-form sum $\hat{Q}_F(T) = (1/8) \sum_{i: T \subseteq \text{vars}(C_i)} \prod_{v \in T} s_{i,v}$ (computable directly from clause structure in $O(m)$ time). Let $r(F) := \|\hat{Q}_F\|_4 / \|\hat{Q}_F\|_2 \in [1, 3^{3/2}]$ be the Bonami-Beckner 4-2 ratio, and set $G(F) := (3^{3/2} - r(F)) / 3^{3/2} \in [0, 1 - 3^{-3/2}]$. Conjecture: there is an absolute constant $c \geq 0.05$ such that for every unsatisfiable 3-CNF F , $\log_2(d^*(F) + 1) \geq c \cdot n \cdot G(F)$; a single F violating this inequality falsifies.

Rationale

Hypercontractive saturation $r(F) \rightarrow 1$ means Q_F has a heavy-tailed degree-3 spectrum: clause-violation behaves like a sharp witness function whose Fourier mass spreads thin, the regime characterizing expander-Tseitin and threshold-random 3-SAT where DPLL must explore exponentially many literal interactions. Conversely $r(F) \rightarrow 3^{3/2}$ (Khintchine-saturated) means Q_F is dominated by a few large coefficients, the regime of structurally simple unsat formulas that DPLL refutes shallowly. The bridge is novel because the Fourier analysis is performed on the CLAUSE STRUCTURE polynomial Q_F (syntactic), not on F 's truth table or its communication matrix, sidestepping the natural-proofs largeness barrier that killed the Cross-Side Fourier attempt.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.18s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df7c90d8.py", line 178, in <module>
    result = run_trial(seed)
             ~~~~~
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df7c90d8.py", line 141, in run_trial
    norm_4 = compute_norm_4(F, n)
             ~~~~~
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df7c90d8.py", line 81, in compute_norm_4
    product *= (1 + s * x[v - 1]) / 2
                ~~~~~
```

IndexError: list index out of range

Judge reasoning

The test crashed with an IndexError in compute_norm_4 before producing any data, so the pre-registered support condition across 480 instances could not be evaluated. | next: Fix the variable indexing bug in compute_norm_4 (likely a 1-based vs 0-based mismatch or variables exceeding n) and rerun the full 480-instance sweep.

Cycle-Type Mixing Defect of Barrington BP Bounds NC^1 Length

- **Verdict:** INCONCLUSIVE
- **Bridge:** Character-theoretic mixing on S_5 — the L^1 distance between the empirical cycle-type distribution of partial products of a deterministic word and the Haar cycle-type distribution $\pi^* = (1, 10, 15, 20, 20, 24, 30)/120$, in the Diaconis-Shahshahani 1981 / Lulov-Pak 2002 / Larsen-Shalev 2008 Plancherel-spectrum tradition. This is a STRUCTURAL invariant of the program (a property of the BP wiring, not of its truth table), and an arXiv search ‘conjugacy

class mixing' AND ('branching program' OR 'Barrington' OR 'formula size') returns 0 direct hits with <5 adjacent papers, all on quantum-circuit Haar approximation. $\times NC^1$ formula leaf size via Barrington 1989 width-5 permutation branching programs over S_5 : every depth- d Boolean formula compiles into a length- 4^d width-5 BP over S_5 whose terminal permutation is the identity iff $f(x)=0$ and the fixed 5-cycle σ_α iff $f(x)=1$. The target separation is NC^1 vs explicit P functions; the structural form of the invariant shields it from the Razborov-Rudich natural-proofs barrier (two formulas with identical truth tables can compile to BPs with very different MIX).

- **Recorded:** 2026-05-18 03:58 UTC
- **Entry ID:** 816a64f0e4e2

Statement

Let $P=(I_1, \dots, I_L)$ be a width-5 BP over S_5 computing a Boolean f via Barrington's protocol, with $I_t=(v_t, \pi_t^{0, \pi_t^{-1}})$. For uniform $X \in \{0,1\}^n$ let $\mu_t^X \in P$ be the distribution on the 7 conjugacy classes of S_5 induced by $\Pi_{\{s \leq t\}} \pi_s^{X_{\{v_s\}}}$, and define $MIX(P) := (1/L) \cdot \sum_{t=1}^L \|\mu_t^{P^{-n}}\|_1$. Conjecture: for every non-constant f and every width-5 Barrington BP P computing f via the standard Ben-Or-Cleve commutator gadget, $MIX(P) \cdot L \geq (1/8) \cdot \log_2 \text{LeafSize}(f)$, where $\text{LeafSize}(f)$ is the minimum literal-leaf count of any AND/OR/NOT formula computing f . A single (P, f) with $MIX(P) \cdot L < (1/8) \cdot \log_2 \text{LeafSize}(f)$ refutes the conjecture.

Rationale

Barrington-compiled BPs realize commutators of 5-cycles whose conjugacy-class trajectory is forced to spend most steps in non-trivial classes when the underlying formula has high algebraic depth; π^* assigns 119/120 mass off the identity class, so the deterministic word can only stay 'near uniform in cycle type' by genuinely commutator-rich computation. Linking $MIX \cdot L$ to $\log \text{LeafSize}$ makes mixing a syntactic complexity certificate that is bypassed by no truth-table proxy, addressing the ACC_SIPSER target of finding natural properties that are STRUCTURAL (not truth-table). The bridge to Lulov-Pak character bounds promises a path to $ACC^0[m]$ via Barrington-Thérien compilation into solvable iterated wreath products without invoking arithmetization (avoiding ALGEBRIZATION).

Novelty

- Judge: NOVEL over 8 arXiv hits

Top hits consulted: - [1709.08890v2] Partial matching width and its application to lower bounds for branching programs - [1107.1826v4] Conjugacy growth of finitely generated groups - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1512.02171v5] Right-jumps and pattern avoiding permutations - [1301.0379v2] The Parameterized Complexity of some Permutation Group Problems

Empirical Test

- exit code: 1, elapsed: 0.12s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 108, in <module>
    result = run_trial(seed)
    ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 85, in run_trial
    metric_value = barrington_protocol(formula, n)
    ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 52, in barrington_protocol
    product *= barrington_protocol(formula[bit], n)
    ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 47, in barrington_pr
    if formula[bit] == 'x':
        ~~~~~^~~~~^
```

IndexError: string index out of range

Judge reasoning

The test crashed with an `IndexError` in `barrington_protocol` before producing any `RESULT` line or data, so neither the support nor falsification condition could be evaluated. | next: Fix the recursive formula parser (handle empty subformulas and properly tokenize literals/operators) and re-run the 30-seed sweep.

NW-Design Restriction Dimension Separates Perm from Padded Det

- **Verdict:** INCONCLUSIVE
- **Bridge:** Nisan-Wigderson combinatorial designs (set systems with bounded pairwise intersection from PRG construction; Nisan-Wigderson 1994; Impagliazzo-Wigderson 1997) $\times$ Mulmuley-Sohoni GCT determinantal complexity $dc(\text{perm}_n)$: orbit-closure separation of perm_n from padded det_m for $m \leq n^{\{1.5\}}$ (Mignon-Ressayre 2004; Landsberg 2017)
- **Recorded:** 2026-05-18 04:25 UTC
- **Entry ID:** 0cac01ce9ab0

Statement

Let $D=(S_1, \dots, S_M)$ be a Nisan-Wigderson design on the index set $[n] \times [n]$ of matrix-entry variables $x_{\{ij\}}$, with each $|S_t|=n$ and pairwise intersections $|S_s \cap S_t| \leq k = \lceil \log_2 n \rceil$. For a homogeneous degree- n polynomial $f \in \mathbb{C}[x_{\{ij\}}]^n$, define the NW-restriction dimension $\rho_D(f) := \dim_{\mathbb{C}} \text{span}\{f^{\wedge\{t\}}(t) : t \in [M]\}$, where $f^{\wedge\{t\}}(t)$ is the polynomial obtained from f by setting $x_{\{ij\}} \mapsto 0$ for every $(i,j) \notin S_t$. CONJECTURE: there exist absolute constants $C_1, C_2 > 0$ with $C_2 > 4C_1 \cdot \log_2 n / 1$ such that for every n and every M with $n \leq M \leq \lfloor n^{\{1.5\}} \rfloor$: (a) for every padded determinant $g(x) = \sum (x)^{\wedge\{n-m\}} \cdot \text{det}_m(L(x))$ with $m \leq \lfloor n^{\{1.5\}} \rfloor$, $\square$ a linear form and L an $m \times m$ matrix of linear forms in the $x_{\{ij\}}$, $\rho_D(g) \leq C_1 \cdot m \cdot \log_2 n$; (b) $\rho_D(\text{perm}_n) \geq C_2 \cdot M$. A single (n, M, D, g) violating (a) or a single (n, M, D) violating (b) refutes the conjecture.

Rationale

NW designs probe disjoint-up-to-log-overlap windows of variables, so the M restrictions act like ‘approximately independent samples’ of the polynomial’s syntactic content. A padded determinant compresses through only $O(m^2)$ hidden linear forms, so its NW-restrictions are forced to live in an $O(m \cdot \text{polylog})$ dimensional subspace; the permanent has no such linear-form bottleneck, and its restrictions should be near-linearly-independent across NW slots. Because ρ_D is an $\exp(n)$ -time syntactic invariant of the polynomial expansion (not of a truth table), it is shielded from Razborov-Rudich natural proofs, and the NW-design layer ties hardness-vs-randomness combinatorics to a GCT obstruction shape distinct from prior Specht/Mahonian/Eulerian attempts.

Novelty

- Judge: NOVEL over 3 arXiv hits

Top hits consulted: - [2202.13016v1] Bounds on Determinantal Complexity of Two Types of Generalized Permanents - [1007.3804v4] Symmetric Determinantal Representation of Formulas and Weakly Skew Circuits - [0106076v3] A determinantal formula for the Hilbert series of one-sided ladder determinantal rings

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b4a3a7c2.py", line 163, in <module>
    result = run_trial(seed)
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b4a3a7c2.py", line 130, in run_trial
    perm_rank = matrix_rank(perm_matrix)
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b4a3a7c2.py", line 33, in matrix_rank
    while pivot < n and matrix[pivot][col] == 0:
    ~~~~~^
```

IndexError: list index out of range

Judge reasoning

The test crashed with an IndexError in matrix_rank before any trials completed, so no data was produced to evaluate the pre-registered support or falsification conditions. | next: Fix the matrix_rank indexing bug (likely a row/column dimension mismatch when building perm_matrix) and rerun the full 270-trial sweep.

Coxeter Inversion Spread of Barrington BPs Bounds NC^1 Depth

- **Verdict:** INCONCLUSIVE
- **Bridge:** Coxeter-group combinatorics of S_5 — the inversion-count / Coxeter length $\square_C: S_5 \rightarrow \{0,1,\dots,10\}$ relative to the simple-transposition generators $s_i=(i,i+1)$, in the Bjorner-Brenti / Humphreys finite-Coxeter-systems tradition (reflection length, weak order, Eulerian-statistics). This corner of finite Coxeter combinatorics is essentially absent from NC^1 / Barrington literature: an arXiv search ‘Coxeter length’ AND (‘Barrington’ OR ‘branching program’ OR ‘ NC^1 ’ OR ‘formula size’) returns 0 direct hits and <5 adjacent papers, none using inversion-count statistics on BP partial-product traces. Distinct from the blacklisted character-mixing attempt: that invariant was a Plancherel/cycle-type Fourier functional, while $\square_C$ is a NON-ring, geometric word-length on S_5 (so it does not extend to polynomial relaxations and cannot algebrize). $\times$ NC^1 formula leaf size via Barrington 1989 width-5 permutation BPs over S_5 : every depth-d Boolean formula compiles into a length- 4^d width-5 BP whose terminal permutation is e iff $f(x)=0$ and a fixed 5-cycle σ iff $f(x)=1$. The structural form of the invariant (a property of the BP wiring, not of the truth table) shields it from the Razborov-Rudich natural-proofs barrier, since two BPs with identical truth tables can yield different σ^2 values.
- **Recorded:** 2026-05-18 04:47 UTC
- **Entry ID:** 66b589aa0f3d

Statement

Let B be a width-5 Barrington BP of length L on n Boolean inputs, with layers $(i_k, g_k^{<0, g_k^{<1}\{k=1..L\}$; for $x \in \{0,1\}^n$ let $\pi_k(x) := g_1^{x_{i_1}} \cdot g_2^{x_{i_2}} \cdot \dots \cdot g_k^{x_{i_k}} \in S_5$ and let $\square_C(\pi)$ be the inversion count of π . Define $\sigma^2(B) := E_{x \sim U(\{0,1\}^n)} [(1/L) \cdot \sum_{k=1}^L (\square_C(\pi_k(x)) - \mu_x)^2]$, where $\mu_x := (1/L) \cdot \sum_k \square_C(\pi_k(x))$. Conjecture: (i) for every BP B obtained by Barrington-compiling a depth- d AND/OR/NOT formula via the canonical Ben-Or-Cleve construction (so $L = 4^d$), $\sigma^2(B) \leq C_1 \cdot d$ for an absolute constant $C_1 \leq 4$; (ii) for every non-constant width-5 BP B (Barrington-compiled or arbitrary), $\sigma^2(B) \geq c_2 > 0$ with $c_2 \geq 1/8$. A single BP violating either bound refutes.

Rationale

Coxeter length is a quantitative geometric measure of how far the partial-product walk wanders inside the Cayley graph $\text{Cay}(S_5, \text{simple transpositions})$; Barrington-compiled BPs traverse highly structured commutator-stamped excursions whose inversion-trace has tightly-controlled fluctuations governed by the recursion depth, while any non-trivial BP must achieve at least one $\square_C > 0$ step and thus a nonzero variance. If true, σ^2 provides a depth-sensitive, truth-table-blind certificate that lower-bounds the formula depth realising a given BP — a structural NC^1 separator immune to natural-proofs and algebrization, since $\square_C$ is not a polynomial functional and does not extend to any commutative-ring relaxation of the Boolean cube.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dee058bd.py", line 84, in <module>
    result = run_trial(seed)
    ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dee058bd.py", line 55, in run_trial
    layers = barrington_compile(int(n**(1/4)))
    ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dee058bd.py", line 30, in barrington_compile
    layers.append(permutation_multiply(sigma, sigma))
    ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dee058bd.py", line 24, in permutation_multiply
    return tuple(p1[p2[i]] for i in range(len(p1)))
    ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dee058bd.py", line 24, in <genexpr>
    return tuple(p1[p2[i]] for i in range(len(p1)))
    ~~~~~^
```

IndexError: tuple index out of range

Judge reasoning

The test crashed with an `IndexError` in `permutation_multiply` before any $\sigma^2(B)$ data was produced, so neither the support nor the falsification condition can be evaluated. | next: Fix the Barrington compiler bug (`permutation_multiply` receiving a malformed tuple from `barrington_compile`) and rerun the $30 \times 4 + 2$ probe protocol.

Birkhoff Spectral Gap of Gate Co-occurrence Bounds AC^0 PARITY Correlation

- **Verdict:** INCONCLUSIVE
- **Bridge:** Birkhoff polytope / doubly-stochastic spectral theory (Sinkhorn 1964; Linial-Samorodnitsky-Wigderson 2000; Forbes-Saxena 2018 on permanental approximations) applied as a STRUCTURAL invariant of the gate-pair input co-occurrence matrix of a Boolean circuit. The $\text{Sym}(n)$ -equivariant Sinkhorn scaling and its λ_2 spectral gap are a non-ring convex-analytic invariant (the scaling map uses divisions and row/column normalisations that do not extend to $\mathbb{F}_q$, so it cannot algebrize). An arXiv search for ‘Sinkhorn scaling’ AND

(‘AC⁰’ OR ‘switching lemma’ OR ‘circuit complexity’) returns 0 direct hits and <5 adjacent papers, all on permanent approximation in algebraic complexity, never on AC⁰ PARITY correlation. × AC⁰[d] circuit correlation with PARITY_n in the Håstad 1986 switching lemma / Razborov–Smolensky regime, in the Cook–Reckhow–Krajíček bounded-arithmetic framework where AC⁰ PARITY size lower bounds correspond to V⁰ ⊆ V⁰(⊕) separations. The invariant is a STRUCTURAL property of the circuit DAG (not of its truth table), shielding it from the Razborov–Rudich natural-proofs barrier; it is Sym(n)-equivariant under the natural action permuting input variables, fulfilling the focus’s ‘natural with respect to a non-trivial group action’ requirement.

- **Recorded:** 2026-05-18 05:16 UTC
- **Entry ID:** 14ed7020b118

Statement

For an AC⁰ circuit C on n inputs of depth d and size s , let $I(g) \subseteq [n]$ be the input cone of gate g (variables reaching g in C). Define $A(C) \in \mathbb{R}^{\{n \times n\} \setminus \{0\}}$ by $A[i,j] = \#\{\text{gates } g : \{x_i, x_j\} \subseteq I(g)\}$, set $\tilde{A} = A + 10^{-3} \cdot J$, and let $D(C)$ be the doubly-stochastic matrix defined by $D(C) = \tilde{A} / \text{tr}(\tilde{A})$. $D(C)$ satisfies $\kappa(C) \leq \exp(-c \cdot n \cdot g(C) / d)$. A single AC⁰ circuit C with $\kappa(C) > \exp(-c \cdot n \cdot g(C) / d)$ (for $c = 1/8$) refutes the conjecture.

Rationale

Hastad’s switching lemma kills AC⁰ correlation with PARITY via random restrictions that spread mass evenly across input coordinates; the Sinkhorn-doubly-stochastic projection captures the SAME spreading geometrically — $g(C)$ measures how ‘mixed’ the gate-pair co-occurrence matrix is under symmetric scaling, and large $g(C)$ forces no input pair to dominate the circuit’s structure. Because $D(C)$ transforms equivariantly under Sym(n) permuting inputs, $g(C)$ is a representation-theoretic invariant of the circuit’s wiring, not of its truth table — so it cannot be a Razborov–Rudich natural property. The construction is purely convex-analytic (Sinkhorn uses divisions, not ring operations) and so does not algebrize under polynomial extensions of the Boolean ring, sidestepping the AVOID examples (tropical homotopy stability, tropical weight accumulation) that hit ALGEBRIZATION.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 5.54s

```
jecture_holds': True, 'counterexample': '', 'n': 16, 'd': 3, 's': 48, 'g': -
0.7359093566514678, 'kappa': 0.0, 'bound': 1.6333060917151618}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.8870418116493577, 'kappa': 0.0, 'bound': 1.8064456904906498}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.8660207193645115, 'kappa': 0.0, 'bound': 1.5418921941310963}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.805622159349614, 'kappa': 0.0, 'bound': 1.4960242355796014}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.9261501129555745, 'kappa': 0.0, 'bound': 1.5889525991259912}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.8006633967206949, 'kappa': 0.0, 'bound': 1.4923196155245912}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
0.7322580901888673, 'kappa': 0.0, 'bound': 2.07977162120796}
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'kappa_bound_ratio', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_
```


Judge reasoning

Every trial reports $\kappa=0.0$ exactly, so the inequality holds vacuously; additionally $g(C)$ is frequently negative, which is impossible for a genuine doubly-stochastic matrix and indicates a Sinkhorn/eigenvalue bug. The pre-registered criterion (≥ 2000 trials, Spearman $\rho \geq 0.3$) is not demonstrated, and the regime tested ($n \leq 16$, tiny random circuits) never approximates PARITY. | next: Fix the Sinkhorn scaling (verify convergence and that $|\lambda_2(D)| \leq 1$), then test on circuits designed to have nontrivial PARI

Mansour L1 Fourier Mass of Clause-Falsification Polynomial Bounds DPLL Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Fourier analysis of Boolean functions — specifically the Linial-Mansour-Nisan / Mansour L^1 Fourier mass $L_1(p) := \sum_S |\hat{p}(S)|$ of low-degree pseudo-Boolean polynomials in the Bonami-Beckner-Mansour tradition; distinct from the 4-norm/2-norm Bonami ratio (black-listed) and from the cross-side symmetric-difference mass (blacklisted), since L_1 is the sum-of-absolute-coefficients functional underlying Mansour’s DNF conjecture and the LMN approximation theorem. An arXiv search for ‘Mansour L1 Fourier mass’ AND (‘Resolution refutation’ OR ‘DPLL’ OR ‘tree-Resolution’) returns 0 direct hits and < 5 adjacent papers, none using L_1 of the clause-count polynomial as a CNF-structural invariant. × Tree-like Resolution / DPLL refutation size $t(F)$ for unsatisfiable 3-CNFs F in the Ben-Sasson-Wigderson / Krajíček bounded-arithmetic regime, where $\neg F$ is Σ^b_0 and $\log_2 t(F)$ lower-bounds V^0 -witnessing complexity; structural shielding from Razborov-Rudich is provided because L_1 depends on the clause LIST (multiplicities, polarities), so two unsat CNFs with identical satisfiability functions (both $\equiv \text{FALSE}$) can have very different L_1 .
- **Recorded:** 2026-05-18 05:56 UTC
- **Entry ID:** 9076d5e87956

Statement

For an unsatisfiable 3-CNF $F = \{C_1, \dots, C_m\}$ on n Boolean variables, let $p_F: \{-1, +1\}^n \rightarrow \mathbb{Z}$, $p_F(x) := \#\{j : x \text{ falsifies } C_j\}$ be the clause-falsification count polynomial, expand it in the Walsh basis $p_F(x) = \sum_S \hat{p}_F(S) \chi_S(x)$, and define the structural Fourier L^1 mass $L_1(F) := \sum_{1 \leq |S| \leq 3} |\hat{p}_F(S)|$. We conjecture that for every unsatisfiable 3-CNF F there is an absolute constant $c \in [0.25, 3.0]$ such that $L_1(F) / \sqrt{m+1} \leq c \cdot \sqrt{(\log_2(t(F) + 1) + 1)}$, where $t(F)$ is the DPLL tree size under the most-frequent-variable branching heuristic; a single unsat F violating the inequality refutes the conjecture.

Rationale

Mansour’s conjecture ($\text{DNF } L_1 \leq s^{\Theta(\log \log s)}$) and the LMN low-degree concentration theorem tie Boolean-function L^1 Fourier mass to small-formula computability; transposing this to CNF refutations predicts that ‘Fourier-spread’ clause sets (large L_1 per $\sqrt{m}$) are forced to have correspondingly deep search trees, since each Walsh monomial in p_F encodes a clause-polarity correlation that any DPLL path must locally certify. The Tseitin family is a sharp test bed: every $\text{XOR}=\omega(v)$ constraint contributes Fourier mass concentrated exactly at one size-3 monomial of magnitude $1/2$ (lower levels cancel by parity), giving $L_1(\text{Tseitin})/\sqrt{m} = \Theta(\sqrt{V})$, parallel to the $\Omega(V)$ Ben-Sasson-Wigderson width lower bound. The invariant is structural (clause-list-dependent), not algebrizing (uses $|\cdot|$ not ring extensions), and not relativizing.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s

```
nces_tested': 0, 'conjecture_holds': False, 'counterexample': 'No valid instances generated'}  
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'No valid instances generated'}  
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'No valid instances generated'}  
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'No valid instances generated'}  
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'No valid instances generated'}  
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'No valid instances generated'}  
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'No valid instances generated'}  
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'No valid instances generated'}  
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'No valid instances generated'}  
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'No valid instances generated'}  
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'No valid instances generated'}  
TRIAL: {'metric_name': 'L1/Fourier_mass_ratio', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'No valid instances generated'}  
RESULT: INCONCLUSIVE reason=no valid instances
```

Judge reasoning

The test generated zero valid instances across all trials (RESULT: INCONCLUSIVE reason=no_valid_instances), so the pre-registered support condition of ≥ 200 instances was not met and no empirical evaluation of $L_1(F)$ or $t^*(F)$ actually occurred. | next: Fix the instance generator to reliably produce unsatisfiable 3-CNFs from the Tseitin, random-density-5, and trivial-unsat families, verify nonzero instances tested, and re-run with the 30-seed protocol.

Chang Spectrum Dimension of Wire-Source DFS Labels Bounds $\text{ACC}^0[2]$ for MOD 3

- **Verdict:** INCONCLUSIVE
- **Bridge:** Additive combinatorics — Chang’s spectral lemma (Chang 2002; Bourgain 2008; Sanders 2019; Green-Ruzsa) bounding the size of the η -spectrum $\text{Spec}_\eta(A) \subset \hat{\mathbb{Z}}/p$ of a set $A \subset \mathbb{Z}/p$ of density α by $O(\log(1/\alpha)/\eta^2)$; applied as a non-ring, Fourier-analytic STRUCTURAL invariant of the circuit DAG (wire-source integer multisets under canonical DFS-from-output labeling), NOT of the truth table. Distinct from the avoided Plünnecke top-Fourier attempt, which used a TRUTH-TABLE Fourier functional; here the Fourier transform is over wire LABELS in $\mathbb{Z}/p$ (gauge fixed by a canonical DFS), so two semantically equivalent circuits yield different χ values, shielding from Razborov-Rudich. The invariant uses $|\cdot|$ and FFT (does not extend to polynomial F_q relaxations), so it does not algebrize per Aaronson-Wigderson. An arXiv search ‘Chang spectral lemma’ AND (‘ACC⁰’ OR ‘circuit complexity’ OR ‘Williams algorithmic method’) returns 0 direct hits and <5 adjacent papers (all on Bohr-set Roth bounds in pure additive combinatorics). $\times$ ACC⁰[2] circuit size for MOD_q on n inputs with q coprime to 2 (focus $q=3$), in the Williams 2011 / Murray-Williams 2018 algorithmic-method regime targeting NEXP $\not\subseteq$ ACC⁰ and explicit-function ACC⁰ lower bounds (Sipser-style candidates).
- **Recorded:** 2026-05-18 06:21 UTC
- **Entry ID:** 29a5335efcbb

Statement

Let C be an $ACC^0[2]$ circuit of size s on n inputs, with gates topologically ordered $g_1, \dots, g_s$ by a canonical DFS-from-output labeling (children visited in lexicographic gate-type then wire-index order). Let p be the smallest prime with $p \geq s+2$. For each MOD_2 gate g of fan-in $k \geq 4$ with input gate-ids $a_1, \dots, a_k \in [s]$, define the spectrum dimension $\chi(g) := |\{\xi \in \{1, \dots, p-1\} : |\sum_{j=1}^k e^{2\pi i a_j \xi/p}| \geq k/2\}|$, and $\chi(C) := (\max_g \chi(g))/\log_2(p+1)$ (taking 0 if C has no MOD_2 gate with fan-in ≥ 4). Conjecture: (i) for every $ACC^0[2]$ circuit C of size s , $\chi(C) \leq C \cdot \log_2(s+1)$ with $C = 8$; and

(ii) every $\text{ACC}^0[2]$ circuit C computing MOD_3 exactly satisfies $\chi(C) \geq C_2 \cdot n / (\log_2(n+2))^2$ with $C_2 = 1/4$. A single random $\text{ACC}^0[2]$ circuit violating (i) or a single MOD_3 -computing $\text{ACC}^0[2]$ circuit violating (ii) refutes the conjecture.

Rationale

Chang's theorem says that the η -spectrum of a random or generic subset of $\mathbb{Z}/p$ is small ($O(\log p)$), so for typical circuits the structural invariant $\chi(C)$ should saturate the upper bound (i). Computing MOD_3 , however, requires the wire-source MOD_2 gates to concentrate Fourier mass at the cube-root-of-unity-like frequencies of $\mathbb{Z}/p$ (because PARITY -composition pathways must reconstruct triadic structure), which forces a Bohr-set-shaped wire-label pattern with large spectrum — giving the lower bound (ii). Because χ depends on the canonical DFS LABELS of the wires (a property of the circuit DAG, NOT of its truth table), the invariant is shielded from Razborov-Rudich, and because it relies on complex-exponential sums (not on ring operations extendable to polynomial relaxations), it is shielded from Aaronson-Wigderson algebrization.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e94115ad.py", line 74, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e94115ad.py", line 51, in run_trial
    circuit = generate_random_acc02_circuit(n, s)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e94115ad.py", line 22, in generate_random_acc02_circuit
    fanin = random.randint(2, min(len(gates) + 1, math.floor(math.log2(s)) + 3))
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/random.py", line 336, in randint
    return self.randrange(a, b+1)
           ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/random.py", line 319, in randrange
    raise ValueError(f"empty range in randrange({start}, {stop})")
ValueError: empty range in randrange(2, 2)
```

Judge reasoning

The test crashed with a `ValueError` in `generate_random_acc02_circuit` before any trials completed, so no data was produced to evaluate either bound (i) or (ii). | next: Fix the fan-in sampling bug (guard $\min(\text{len}(\text{gates})+1, \text{floor}(\log_2(s))+3) \geq 2$, e.g., skip MOD gate creation until enough predecessor gates exist) and rerun the 360 random instances plus the MOD_3 enumeration.

Bourgain Noise-Sensitivity Mass of Clause-Falsification Polynomial Lower-Bounds DPLL Size

- **Verdict:** INCONCLUSIVE

- **Bridge:** Bourgain noise-sensitivity / sharp-threshold Fourier analysis — the high-frequency Fourier functional $\text{NSM}(f) = \sum_S (1 - \rho^{|S|}) \cdot \hat{f}(S)^2$ as a structural invariant, in the Bourgain 1999 / Kahn-Kalai-Linial 1988 / Friedgut-Kalai 1996 / Mossel-O'Donnell-Oleszkiewicz 2010 tradition. Distinct from Mansour-L1 ($\sum_S |\hat{f}(S)|$, blacklisted), the Bonami 4-norm/2-norm ratio (blacklisted), and the cross-side symmetric-difference mass (blacklisted, NATURAL_PROOFS-hit on a truth-table Fourier object): NSM is a quadratic, level-weighted Fourier mass that captures Bourgain-style ‘sensitivity at scale $1/n$ ’ rather than absolute coefficient sums. The functional uses absolute values and a level-indexed weighting $\sum_S (1 - \rho^{|S|})$ that does not extend to a polynomial F_q relaxation, so the bridge does not algebrize per Aaronson-Wigderson. An arXiv search for ‘Bourgain noise sensitivity’ AND (‘DPLL’ OR ‘tree-Resolution’ OR ‘CNF refutation’) returns 0 direct hits and <5 adjacent papers (none using NSM on the syntactic clause-falsification polynomial as a CNF invariant). × Tree-like Resolution / DPLL refutation size $t(F)$ for unsatisfiable 3-CNFs F in the Ben-Sasson-Wigderson / Krajíček bounded-arithmetic regime where $\neg F$ is Σ^b_0 and $\log_2 t(F)$ lower-bounds V^0 -witnessing complexity. Structural shielding from Razborov-Rudich is provided because $\text{NSM}(F)$ is a functional of the clause LIST (multiplicities, polarities) — two unsat CNFs with identical satisfiability functions (both $\equiv \text{FALSE}$) can have very different $\text{NSM}(F)$ — so the invariant is not a poly-time-computable property of truth tables.
- **Recorded:** 2026-05-18 06:47 UTC
- **Entry ID:** 2f957f34a9d5

Statement

For every unsatisfiable 3-CNF F on $n \geq 10$ variables with m clauses, define $\psi_F: \{-1, +1\}^n \rightarrow [0, 1]$ by $\psi_F(x) = (1/m) \cdot \#\{C \in F : C \text{ is falsified by } x\}$, with Fourier expansion $\psi_F = \sum_S \hat{\psi}_F(S) \cdot \chi_S$ under uniform measure. Set $\rho = 1 - 1/n$ and define $\text{NSM}(F) := (\text{Var}(\psi_F) - \text{Stab}_\rho(\psi_F)) / \text{Var}(\psi_F) = \sum_{S \neq \emptyset} (1 - \rho^{|S|}) \cdot \hat{\psi}_F(S)^2 / \sum_{S \neq \emptyset} \hat{\psi}_F(S)^2 \in [0, 1]$. Conjecture: there is an absolute constant $c \geq 1/16$ such that $\log_2 t(F) \geq c \cdot n \cdot \text{NSM}(F)$, where $t(F)$ is the size of the canonical tree-Resolution refutation produced by a fixed deterministic DPLL with unit propagation and lexicographic branching; a single unsat 3-CNF violating this inequality refutes the conjecture.

Rationale

Bourgain’s noise-sensitivity machinery isolates ‘high-frequency’ Fourier mass of structural polynomials; for an unsat CNF, large $\text{NSM}(F)$ means ψ_F ’s Fourier energy sits at levels $\Omega(n)$, i.e. the falsification pattern depends on $\Theta(n)$ -tuples of variables rather than a small junta. Because DPLL’s branching corresponds to restriction operators on ψ_F and a small refutation tree exhibits a low-junta restriction-tree for ψ_F (each leaf restricts ψ_F to a single falsified clause), high noise-sensitivity should force exponentially many leaves — paralleling how KKL/Bourgain bound junta-approximation by noise sensitivity. The invariant is syntactic (clause-list dependent), quadratic in Fourier coefficients with a non-ring level-weighting, and computable in $2^n \cdot n$ time, so it sidesteps Natural Proofs, Algebrization, and Relativization simultaneously.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.07s

No unsat instances found}

```
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False}
```

```

TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False}
TRIAL: {'metric_name': 'margin', 'metric_value': 0, 'instances_tested': 0, 'conjecture_holds': False}
RESULT: INCONCLUSIVE reason=insufficient_support

```

Judge reasoning

The test generator produced zero unsatisfiable 3-CNF instances across all trials, and the RESULT line explicitly reports INCONCLUSIVE due to insufficient support. The pre-registered threshold of ≥ 240 accepted instances was not even approached (0 tested). | next: Replace the random-generation pipeline with a guaranteed-unsat construction (e.g., random 3-CNF at $\alpha \approx 4.6$ filtered by a SAT solver, or structured families like pigeonhole/Tseitin encoded as 3-CNF) to ensure ≥ 240 accepted instances before

Generated Algebra Dimension of Layer Operators Bounds IP₂ Read-Twice BP

- **Verdict:** INCONCLUSIVE
- **Bridge:** Operator algebra / Burnside-Schur-Wedderburn theory — the unital R -subalgebra $A(P) \subseteq R^{\{w \times w\}}$ generated by the layer transition matrices $\{T_i^{\{b\}}\}$ of a branching program P , with $\dim A(P)$ computed by iterative span closure (start from identity and all $T_i^{\{b\}}$ flattened to $R^{\{w^2\}}$; multiply each existing basis element by each layer operator, add the product to the span via Gauss-Jordan, iterate until no new dimension is found). Distinct from prior free-probability attempts (S-transform, Laplacian R -transform, free cumulants on Max-Cut) which targeted communication / SDP gaps; here the algebra invariant is a STRUCTURAL property of the BP wiring (not its truth table — two BPs computing the same function can have very different $\dim A$), shielding from Razborov-Rudich. The closure uses real linear-algebra rank operations (not ring identities over F_q), shielding from Aaronson-Wigderson algebrization. An arXiv search ‘generated matrix algebra’ AND (‘branching program’ OR ‘read-twice’ OR ‘Borodin Razborov Smolensky’) returns 0 direct hits with < 5 adjacent papers (all on Barrington permutation BPs over a fixed simple group algebra, never on the generated algebra of arbitrary read-twice transitions). $\times$ Read-twice oblivious BP size $s(P)$ for $IP_2(x,y) = XOR_k(x_k \text{ AND } y_k)$ on $2n$ inputs under the adversarial variable order $x_1, \dots, x_n$ followed by $y_1, \dots, y_n$ (Borodin-Razborov-Smolensky 1993 regime); instantiating the focus’s ‘distinguishing tensor’ formalism: find $\rho(P)$ with $\rho(P) = O(\log \text{size}(P))$ but $\rho(P_{IP}) = \Omega(n)$ for any P computing IP_2 .
- **Recorded:** 2026-05-18 07:28 UTC
- **Entry ID:** 59a86c7a7d5c

Statement

Let P be a read-twice oblivious BP on $2n$ inputs with width w , length L , and $2L$ layer transition matrices $\{T_i^{\{b\}} \in \{0,1\}^{\{w \times w\}} : i \in [L], b \in \{0,1\}\}$ where each row of $T_i^{\{b\}}$ has at most one 1. Let $A(P) \subseteq R^{\{w \times w\}}$ be the unital R -subalgebra generated by the identity and the $\{T_i^{\{b\}}\}$, and define $\rho(P) := \log_2(\dim_R A(P) + 1)$. Conjecture: (i) for every read-twice oblivious BP P , $\rho(P) \leq 2 \cdot \log_2(w + 1)$ (trivially since $\dim A(P) \leq w^2 \leq \text{size}(P)^2$); (ii) for every read-twice oblivious BP P computing IP_2 under the adversarial input order, $\rho(P) \geq n - 2$. A single read-twice oblivious BP P with $f_P = IP_2$ (adversarial order) and $\rho(P) < n - 2$ refutes (ii); a single BP violating (i) refutes (i).

Rationale

A(P) encodes EVERY product of layer transitions, i.e. all reachable state-permuters of the BP's syntactic dynamical system — a relational/categorical span, not a single truth-table number, so it dodges the textbook Razborov-Rudich trap and the recent calibration warnings against truth-table single-number invariants. By the BRS rectangle argument any read-once BP for IP₂ in adversarial order must distinguish 2^n x-prefixes at the midpoint cut; we conjecture this distinguishing power forces a Schur-Wedderburn-large generated algebra even with two reads per variable, because the y-phase must non-trivially recombine all 2^n prefix-states into a single parity bit. The span-closure construction uses real rank operations only, so it does not extend to a polynomial F_q relaxation and is shielded from Aaronson-Wigderson algebrization.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e6d0879e.py", line 158, in <module>
    main()
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e6d0879e.py", line 140, in main
    result = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e6d0879e.py", line 123, in run_trial
    rho = compute_rho(P, var_positions, w)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e6d0879e.py", line 110, in compute_rho
    dimension = compute_dimension(basis)
                ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e6d0879e.py", line 60, in compute_dimension
    basis = gaussian_elimination(basis)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e6d0879e.py", line 52, in gaussian_elimination
    matrix[k][j] -= factor * matrix[i][j]
    ~~~~~^
```

IndexError: list index out of range

Judge reasoning

The test crashed with an IndexError inside gaussian_elimination before producing any RESULT line or data, so neither the support nor falsification condition can be evaluated. | next: Fix the Gaussian elimination routine (ensure consistent row/column dimensions and bounds checks) and rerun the 360 random + canonical/friendly IP₂ trials.

Szegedy Quantum-Walk Phase Gap Lower-Bounds Tseitin Tree-Resolution

- **Verdict:** INCONCLUSIVE
- **Bridge:** Discrete-time Szegedy / Grover quantum walks on finite graphs (Szegedy 2004 'Quantum speed-up of Markov chain based algorithms'; Magniez-Nayak-Roland-Santha 2011 'Search via quantum walk'; Krovi-Magniez-Ozols-Roland 2016) — the phase gap $\Delta_Q(G)$ of the Szegedy walk operator $W=(2\Pi_B-I)(2\Pi_A-I)$ acting on the edge-orientation space $C^{\{2m\}}$, accessed through the Szegedy correspondence $\Delta_Q = 2 \cdot \arcsin(\sqrt{\text{gap}(P_G)})$ where P_G is the

simple-random-walk transition matrix. This corner of quantum-walk operator theory is essentially absent from proof complexity: an arXiv search ‘Szegedy quantum walk’ AND (‘Tseitin’ OR ‘resolution refutation’ OR ‘DPLL’ OR ‘proof complexity’) returns 0 direct hits, with <5 adjacent papers (all on quantum mixing-time speedups, never on CNF-refutation lower bounds). Distinct from the blacklisted magnetic-Laplacian holonomy attempt: that invariant was the smallest gauge-invariant eigenvalue of a signed Bochner operator on 0-forms, while Δ_Q is a phase-gap on the EDGE-ORIENTATION (line-graph-like) space, computed from singular values of the discriminant operator rather than gauge-fixed signed spectra. × Tree-like Resolution / DPLL refutation size $t(T(G,\omega))$ for Tseitin XOR formulas $T(G,\omega)$ on connected 3-regular graphs G with odd charge ω (Urquhart 1987 / Ben-Sasson-Wigderson 2001 / Alekhnovich-Razborov 2001 expansion regime), in the Cook-Reckhow-Krajíček bounded-arithmetic framework where $\neg T(G,\omega)$ is Σ^b_0 and $\log_2 t$ lower-bounds V^0 -witnessing complexity.

- **Recorded:** 2026-05-18 08:08 UTC
- **Entry ID:** 97906ec529b6

Statement

For a connected 3-regular graph G on n vertices with odd charge ω , define $\nu(G) := \lfloor n \cdot \Delta_Q(G) \rfloor$ where $\Delta_Q(G) = 2 \cdot \arcsin(\sqrt{1 - |\lambda_2(P_G)|})$ is the Szegedy quantum-walk phase gap on G ’s edge-orientation space and $\lambda_2(P_G)$ is the second-largest-magnitude eigenvalue of the simple-random-walk matrix $P_G = D^{-1}A$. Then there exist absolute constants $c_1, c_2 > 0$ such that (i) every 3-regular G with min balanced edge-bisection $\leq \sqrt{n}$ satisfies $\nu(G) \leq c_1$, and (ii) for every odd ω , $t(T(G,\omega)) \geq 2^{(c_2 \cdot \nu(G))}$. A single 3-regular G with odd ω and $t(T(G,\omega)) < 2^{(\nu(G)/8)}$ refutes (ii); a single G with bisection $\leq \sqrt{n}$ and $\nu(G) > 16$ refutes (i).

Rationale

The Szegedy walk’s phase gap encodes a topological obstruction to coherent superposition mixing across edge ORIENTATIONS — exactly the structure that Tseitin XOR formulas force a Resolution refutation to track via the F_2 -cycle-space mechanism (Urquhart-BSW). Because Δ_Q is built on the line-graph edge space rather than the vertex Laplacian, it captures orientation-sensitive bottlenecks invisible to plain vertex spectral gap, while remaining poly-time computable via one symmetric eigendecomposition. The invariant is shielded from Razborov-Rudich (structural property of G , not of any truth table) and from Aaronson-Wigderson (the $\arcsin/\sqrt{}$ operations do not extend to a polynomial F_q relaxation).

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_086fec4c.py", line 225, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_086fec4c.py", line 198, in run_trial
    lambda_2 = compute_second_largest_eigenvalue(P)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_086fec4c.py", line 138, in compute_second_largest_eigenvalue
    eigenvalues = matrix_eigenvalues(P)
                ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_086fec4c.py", line 91, in matrix_eigenvalues
    raise ValueError("Matrix too large for this implementation")
```

ValueError: Matrix too large for this implementation

Judge reasoning

The test crashed with a ValueError ('Matrix too large for this implementation') in the eigenvalue computation before any of the 270 instances could be evaluated, so neither support nor falsification conditions were tested. | next: Replace the custom matrix_eigenvalues routine with numpy.linalg.eig or scipy.sparse.linalg.eigs to handle the $2n \times 2n$ matrices for $n \in \{12, 16, 20\}$, then rerun the 270-instance protocol.

Halász L^2 Spectrum Discrepancy Lower-Bounds Sign-Matrix Rigidity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Erdős-Turán / Halász L^2 discrepancy of empirical measures on $[0,1]$ (Halász 1981 'On the boundedness of digital nets discrepancy'; Drmota-Tichy 1997 ch.1; Vaaler 1985 majorant-minorant; Niederreiter-Pillichshammer QMC tradition) — a non-ring, integral-of-CDF-deviation functional that uses sup/abs/integration and does NOT extend to polynomial relaxations over F_q (Aaronson-Wigderson algebrization-safe). Applied here as the L^2 deviation of the rescaled empirical SQUARED-singular-value CDF of a sign matrix against the maximum-entropy (uniform-on-spectrum) reference. An arXiv search 'Halász discrepancy' AND ('matrix rigidity' OR 'Valiant rigidity') returns 0 direct hits; <5 adjacent papers on QMC discrepancy vs spectral concentration of random ± 1 ensembles, none targeting rigidity. Distinct from blacklisted Bourgain-Tzafriri sub-column dispersion (column-subset operator norms) and signed-support L_∞ discrepancy (Beck-Fiala on incidence rows). $\times$ Matrix rigidity $R_M(r)$ for sign matrices $M \in \{\pm 1\}^{N \times N}$ in the Valiant 1977 / Friedman 1993 / Razborov 1989 / Lokam 2009 regime, accessed through the Friedman spectral lower bound $R_F(M, r) := \sigma_{r+1}(M)^2 \leq R_M(r)$. Target separation: unsaturated rigidity bounds for explicit sign families (Hadamard, padded-identity, low-rank-plus-noise), the canonical stepping stone toward log-depth arithmetic-circuit lower bounds for linear maps via the Valiant rigidity program.
- **Recorded:** 2026-05-18 08:28 UTC
- **Entry ID:** 618e9867e153

Statement

Let $M \in \{\pm 1\}^{N \times N}$ have singular values $\sigma_1 \geq \dots \geq \sigma_N \geq 0$; since $\|M\|_{F^2}^2 = N^2$ the weights $p_i := \sigma_i^2 / N^2$ form a probability distribution on $[N]$. Define the empirical descending-cumulative CDF $F_M: [0,1] \rightarrow [0,1]$ by $F_M(t) := \sum_{i \leq [tN]} p_i$ and the Halász L^2 discrepancy against the uniform-spectrum reference $\bar{U}(t) = t$ by $D_2(M) := (\int_0^1 (F_M(t) - t)^2 dt)^{1/2} \in [0, 1/\sqrt{3}]$. Conjecture: there exists an absolute constant $c > 0$ such that for every $M \in \{\pm 1\}^{N \times N}$, $\sigma_{\lfloor N/2 \rfloor + 1}(M)^2 \geq c \cdot N \cdot (1 - 4D_2(M))_+$ (where $(x)_+ := \max(0, x)$); a single sign matrix exhibiting $\sigma_{\lfloor N/2 \rfloor + 1}^2 < (c/2) \cdot N \cdot (1 - 4D_2)_+$ falsifies it.

Rationale

Halász-type L^2 discrepancy measures how concentrated the squared spectrum is relative to the entropy-maximizing flat profile; a flat (low-discrepancy) spectrum forces $\sigma_{\lfloor N/2 \rfloor + 1}$ to remain $\Theta(\sqrt{N})$, the canonical Friedman regime where the rigidity lower bound is tight. The bridge encodes the heuristic that 'spectral-uniformity $\Rightarrow$ rigidity', upgraded to a quantitative discrepancy-to-Friedman bound, and the use of integration plus absolute values blocks any polynomial-extension lift that would trigger algebrization. Because D_2 is computed from M itself (not from a Boolean truth-table predicate of the function it computes), the bound is structural rather than natural-proofs-style.

Novelty

- Judge: NOVEL over 8 arXiv hits

Top hits consulted: - [1007.3178v1] Combination of CDF and D0 Results on the Mass of the Top Quark - [1107.5255v3] Combination of CDF and DO results on the mass of the top quark using up to 5.8~fb⁻¹ of data - [9810031v1] Search for a Technicolor omega_T Particle in Events with a Photon and a b-quark Jet at CDF - [2207.14810v3] Simplifying a classical-quantum algorithm interpolation with quantum singular value transformations - [1602.03155v2] From K.A.M. Tori to Isospectral Invariants and Spectral Rigidity of Billiard Tables

Empirical Test

- exit code: 0, elapsed: 113.81s

```
3, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.0503420448340177, "instances_t
TRIAL: {"seed": 503, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.047414478541
TRIAL: {"seed": 547, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.046467970249
TRIAL: {"seed": 593, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.043455103440
TRIAL: {"seed": 631, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.044218604171
TRIAL: {"seed": 677, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.046363047660
TRIAL: {"seed": 727, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.043441508690
TRIAL: {"seed": 773, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.044789497691
TRIAL: {"seed": 821, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.042519796501
TRIAL: {"seed": 877, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.044574638238
TRIAL: {"seed": 929, "metric_name": "sigma_{N/2+1}^2 / (N*(1-4D_2))", "metric_value": 1.041993238564
RESULT: SUPPORTED mean=1.0448 std=0.0022 support_fraction=1.0000
```

Judge reasoning

The test only samples uniformly random ± 1 matrices, whose spectra concentrate tightly around Marchenko–Pastur (ρ clusters at 1.044 ± 0.002 , $\text{SNR} \approx 474$), so the adversarial/structured sign matrices relevant to rigidity are never probed; the near-vacuous regime where $(1-4D_2)_+$ is small-but-positive is also untested. | next: Re-run including structured constructions (Hadamard, Sylvester, circulant ± 1 , low-rank-perturbed sign matrices, and Paley/quadratic-residue matrices) plus targeted instances engi

Mostar Index of Gate-Adjacency Graph Bounds $\text{ACC}^0[m]$ Size for MOD_q

- **Verdict:** INCONCLUSIVE
- **Bridge:** Extremal chemical graph theory — the Mostar index $\text{Mo}(G) = \sum_{uv \in E(G)} |n_u(uv) - n_v(uv)|$, where $n_u(uv)$ counts vertices strictly closer to u than to v (Došlić–Martinjak–Škrekovski–Spuženić–Tepoh 2018 ‘Mostar index’; Tratnik 2019; Arockiaraj–Klavžar 2020; Hayat–Liu 2021). $\text{Mo}(G)$ is a structural distance-balance invariant of finite graphs, computed from all-pairs shortest paths and absolute differences — non-ring operations that do NOT extend to polynomial relaxations over $\mathbb{F}_q$, shielding the invariant from Aaronson–Wigderson algebrization. The bridge to circuit complexity has essentially zero presence in the literature: an arXiv search for ‘Mostar index’ AND (‘ ACC^0 ’ OR ‘circuit complexity’ OR ‘Williams algorithmic method’ OR ‘Boolean circuit’) returns 0 direct hits and < 5 adjacent papers, all on chemical isomer enumeration. Distinct from blacklisted Möbius/independence/Birkhoff/Plünnecke attempts: Mo measures edge-wise distance asymmetry on the circuit DAG’s underlying graph, not poset incidence values, simplicial Euler characteristics, doubly-stochastic Sinkhorn limits, or top-Fourier doubling. $\times \text{ACC}^0[m]$ circuit size for MOD_q on n inputs with q coprime to m (focus $q=3$, $m=2$), in the Williams 2011 / Murray–Williams 2018 algorithmic-method regime targeting $\text{NEXP} \not\subseteq \text{ACC}^0$ and explicit-function ACC^0 lower bounds. The invariant

is a STRUCTURAL property of the circuit DAG (two $\text{ACC}^0[m]$ circuits computing the same function can have very different Mo values), shielding from Razborov–Rudich natural-proofs.

- **Recorded:** 2026-05-18 08:56 UTC
- **Entry ID:** b7ec55b48398

Statement

For an $\text{ACC}^0[m]$ circuit C with s gates and depth d , let G_C be the underlying undirected adjacency multigraph on gate vertices (with each wire contributing one edge) and let $\text{Mo}(C) := \sum_{uv \in E(G_C)} |n_u(uv) - n_v(uv)|$ denote its Mostar index, with $n_u(uv) = |\{w : d_{G_C}(w,u) < d_{G_C}(w,v)\}|$. Conjecture: (i) for every $\text{ACC}^0[m]$ circuit C of size s and depth d , $\text{Mo}(C) \leq C_1 \cdot s^2 \cdot d$ for an absolute constant $C_1 \leq 4$; (ii) for every $\text{ACC}^0[2]$ circuit C computing MOD_3 on n inputs ($n \geq 5$), $\text{Mo}(C) \geq C_2 \cdot n \cdot s^{\{1/2\}}$ for $C_2 \geq 1/16$. A single $\text{ACC}^0[2]$ circuit violating (i), or a MOD_3 $\text{ACC}^0[2]$ circuit violating (ii), falsifies the conjecture.

Rationale

Williams-style structural exploitations succeed by extracting circuit-wiring properties that are necessarily small for fast circuit-SAT yet large on hard symmetric functions; $\text{Mo}(C)$ measures cumulative distance-balance asymmetry, which is forced to be tame on shallow, locally-concentrated ACC wirings (clause i) but forced to spread on circuits that must reach every input to compute a fully-symmetric function like MOD_3 (clause ii). The invariant uses no ring operations and is not a truth-table functional, so it evades both algebrization and natural-proofs. The distance-balance angle is genuinely new to circuit complexity — Mostar’s recent introduction (2018) has kept it confined to chemical graph theory.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [0711.2360v2] Every longest circuit of a 3-connected, $K_{3,3}$ -minor free graph has a chord - [1012.3942v1] Distance-balanced closure of some graphs

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fed4589a.py", line 160, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fed4589a.py", line 84, in run_trial
    gates, adj = generate_acc0_circuit(seed, s, d)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fed4589a.py", line 53, in generate_acc0
    inputs.append(random.randint(0, n-1))
                   ^^^^^^^^^^^^^^^^^
File "/usr/lib/python3.12/random.py", line 336, in randint
    return self.randrange(a, b+1)
           ^^^^^^^^^^^^^^^^^
File "/usr/lib/python3.12/random.py", line 319, in randrange
    raise ValueError(f"empty range in randrange({start}, {stop})")
ValueError: empty range in randrange(0, 0)
```

Judge reasoning

The test crashed with a ValueError in generate_acc0_circuit (empty range in randrange(0, 0)) before producing any data, so neither support nor falsification can be assessed. | next: Fix the circuit generator to guard against n=0 (ensure input count ≥ 1 before sampling) and rerun the 360 random + $\geq 20 \text{ MOD } 3$ instance battery.

Patience-Sort LIS of XOR-Lifted Row Permutations Bounds CC^D

- **Verdict:** INCONCLUSIVE
- **Bridge:** Robinson-Schensted-Knuth / Vershik-Kerov Plancherel limit-shape combinatorics — the longest-increasing-subsequence functional $\lambda_1(w)$ on integer words, computed in $O(|w| \log |w|)$ by patience sorting (Knuth 1970; Logan-Shepp 1977; Vershik-Kerov 1977; Baik-Deift-Johansson 1999 Tracy-Widom fluctuations; Romik 2015). An arXiv search for ‘longest increasing subsequence’ AND (‘lifting theorem’ OR ‘communication complexity’ OR ‘XOR-function’) returns 0 direct hits and <5 adjacent papers (Plancherel-measure work in random-matrix theory, never on lifted communication matrices). The LIS functional uses only comparison/sort (a non-ring partial-order operation), so it does NOT extend to a polynomial extension of the Boolean ring and is safe from Aaronson-Wigderson algebrization. Distinct from blacklisted Tamari/Palio rank (a tree-order statistic), Gromov 4-point δ (a metric invariant), and Burau trace (a representation-theoretic invariant). × Deterministic two-party communication complexity of XOR-gadget-lifted Boolean functions $M_{\{f \cdot \text{XOR}_k\}}(a,b)=f(a \oplus b)$, in the Raz-McKenzie / Göös-Pitassi-Watson / Hatami-Hosseini-Lovett lifting framework where CC^D of XOR-lifts is the canonical Log-Rank Conjecture testbed and the prototypical query-to-communication target.
- **Recorded:** 2026-05-18 09:23 UTC
- **Entry ID:** dd4c952aa3ea

Statement

For Boolean $f: \{0,1\}^k \rightarrow \{0,1\}$ with on-set $S(f)=f^{-1}(1)$, $|S(f)| \geq 2$, define for each $a \in \{0,1\}^k$ the row-permutation word $\tau_a = (b \oplus a : b \in S(f), b \text{ listed in lex order})$, and set $\mu(f) = \max_{a \in \{0,1\}^k} \text{LIS}(\tau_a)$, where LIS is the strictly-increasing-subsequence length via patience sorting. Conjecture: for every such f , $\mu(f) \leq \text{rank}_{\{F_2\}}(M_{\{f \cdot \text{XOR}_k\}}) + 1$, and consequently $\lceil \log_2 \mu(f) \rceil \leq CC^D(M_{\{f \cdot \text{XOR}_k\}}) + 1$ via the canonical algebraic lower bound. A single Boolean f with $\mu(f) \geq \text{rank}_{\{F_2\}}(M_{\{f \cdot \text{XOR}_k\}}) + 2$ refutes the conjecture.

Rationale

XOR-lifting embeds the additive $(F_2^k, \oplus)$ structure of f into a $2^{k \times 2}k$ communication matrix whose rows are shifts of the indicator of $\bar{S}(f)$. Vershik-Kerov asserts that random permutations of length n have $\text{LIS} \sim 2\sqrt{n}$; structured shifts (affine $S(f)$, low-rank M) align entries into long monotone runs (large LIS), while ‘rigid’ Boolean functions (e.g., inner product, parity-twist) yield close-to-random permutations with bounded LIS. Patience sorting probes precisely the increasing-monotone substructure that F_2 -rank misses linearly, so a non-trivial μ -vs-rank inequality would expose new XOR-lifting structure — and any counterexample is a witness to RSK NOT respecting the XOR-orbit structure, falsifying the conjecture in one shot.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.11s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a979fd53.py", line 111, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a979fd53.py", line 73, in run_trial
    M_all.append(M)
                ^
```

UnboundLocalError: cannot access local variable 'M' where it is not associated with a value

Judge reasoning

The test crashed with an UnboundLocalError before producing any data, so no instances were evaluated and the pre-registered support condition cannot be assessed. | next: Fix the UnboundLocalError around line 73 (ensure M is always assigned before append, likely by initializing M or handling the branch where it isn't computed) and rerun the full 30 seeds $\times$ $k \in \{3,4,5\}$ sweep.

Effective Resistance of Charged Vertex Pairs Bounds Tseitin DPLL Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Electrical network theory / effective-resistance metric — the symmetric kernel $R_{\text{eff}}(u,v) = L^+(u,u) + L^+(v,v) - 2 \cdot L^+(u,v)$ computed from the Moore-Penrose pseudoinverse of the combinatorial Laplacian (Doyle-Snell 1984 'Random Walks and Electric Networks'; Klein-Randić 1993 'Resistance distance'; Chandra-Raghavan-Ruzzo-Smolensky-Tiwari 1996 'The electrical resistance of a graph captures its commute and cover times'; Spielman-Srivastava 2011 graph sparsification). This is a non-ring, real-analytic GRAPH METRIC on $V \times V$ — the pseudoinverse uses real linear algebra and admits NO polynomial extension to F_q (the Sherman-Morrison-style identity for L^+ fails over finite rings), so the invariant is safe from Aaronson-Wigderson algebrization. Distinct from blacklisted Tseitin attempts: 2-Sylow rank of the critical group is an integer-cokernel invariant; the signed-Laplacian holonomy is a gauge-fixed Bochner eigenvalue on a $\mathbb{Z}/2$ line bundle; the Szegedy quantum-walk phase gap is a singular-value on the edge-orientation space. R_{eff} is none of these — it is a pseudoinverse-diagonal pairing weighted by the charge support. An arXiv search 'effective resistance' AND ('Tseitin' OR 'resolution refutation' OR 'proof complexity') returns 0 direct hits with <5 adjacent papers (Spielman-style sparsifier/pseudorandomness work, never as a CNF refutation lower bound). $\times$ Tree-like Resolution / DPLL refutation size $t(T(G,\omega))$ for Tseitin XOR formulas $T(G,\omega)$ on connected 3-regular graphs G with odd charge ω (Urquhart 1987 / Ben-Sasson-Wigderson 2001 / Alekhnovich-Razborov 2001 expander regime), in the Cook-Reckhow-Krajíček bounded-arithmetic framework where $\neg T(G,\omega)$ is Σ^b_0 and $\log_2 t$ lower-bounds V^0 -witnessing complexity. The invariant is STRUCTURAL on (G,ω) (two (G,ω) pairs computing the same constant-FALSE function differ in ρ), shielding from Razborov-Rudich natural proofs.
- **Recorded:** 2026-05-18 09:54 UTC
- **Entry ID:** 0f748361fa5f

Statement

For every connected 3-regular graph G on n vertices and every odd-parity charge $\omega \in F_2^V$, define the charged-pair resistance $\rho(G,\omega) := (1/n) \cdot \sum_{u \neq v: \omega(u)=\omega(v)=1} R_{\text{eff}_G}(u,v)$ when $|\text{supp}(\omega)| \geq 2$, and $\rho(G,\omega) := (1/n) \cdot \sum_{v: \omega(v)=0} R_{\text{eff}_G}(v_0, v)$ when $\text{supp}(\omega)=\{v_0\}$, with R_{eff_G} obtained from the Moore-Penrose pseudoinverse of the graph Laplacian. Conjecture: there is an absolute constant $c \geq 1/200$ such that for every such (G,ω) , $\log_2 t(T(G,\omega)) \geq c \cdot n \cdot \rho(G,\omega)$; equivalently no DPLL search tree certifies $\neg T(G,\omega)$ with fewer than $2^{c \cdot n \cdot \rho(G,\omega)}$ leaves. A single (G,ω) with $n \leq 40$ satisfying $\log_2 t(T(G,\omega)) < 0.005 \cdot n \cdot \rho(G,\omega)$ refutes the conjecture.

Rationale

Effective resistance is the canonical electrical witness of bottlenecks: $R_{\text{eff}}(u,v)$ is small exactly when many edge-disjoint paths connect u and v with low congestion, which is precisely the regime where parity-flow between charged vertices can be locally cancelled — and so where Tseitin should be EASY; conversely large ρ encodes electrical isolation between the parity defects ω carries, which we conjecture forces DPLL to enumerate many sub-branches to certify a global obstruction. The bridge to proof complexity is novel because the standard expander-Tseitin lower bound goes through cycle-space-mod-2 + Ben-Sasson-Wigderson width, never through Doyle-Snell electrical flows; coupling resistance to refutation size would give a quantitative, non-spectral, non-algebraic lower-bound certificate that does not extend to any polynomial F_q relaxation, sidestepping algebraization barriers.

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [3506702] On Proof Complexity of Resolution over Polynomial Calculus - [LIPIcs.MFCS.2019.49] Bounded-depth Frege complexity of Tseitin formulas for all graphs

Empirical Test

- exit code: 1, elapsed: 0.12s

```
'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'math d
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c
TRIAL: {'metric_name': 'R', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, 'c
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e646b8c2.py", line 253, in <module>
    trial = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e646b8c2.py", line 204, in run_trial
    adj = generate_3_regular_graph(n, seed + _)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e646b8c2.py", line 25, in generate_3_re
    v = random.choice([x for x in stubs if x != u])
            ~~~~~
```

```
File "/usr/lib/python3.12/random.py", line 347, in choice
    raise IndexError('Cannot choose from an empty sequence')
```

IndexError: Cannot choose from an empty sequence

Judge reasoning

The test crashed with an `IndexError` in the 3-regular graph generator (empty stub sequence) and only produced ‘math domain error’ counterexamples with `metric_value` 0.0, so no valid DPLL-terminating instances were collected to evaluate the pre-registered support or falsification conditions. | next: Fix the 3-regular graph sampler (e.g., use a configuration-model with retry/rejection or `networkx.random_regular_graph`) and guard ρ computation against degenerate supports before rerunning the 180-insta

p=3 Path-Family Modulus Lower-Bounds Tseitin Tree-Resolution

- **Verdict:** INCONCLUSIVE
- **Bridge:** Discrete quasiconformal analysis — p-modulus of path families on weighted graphs for $p \neq 2$ (Beurling 1946 modulus tradition; Heinonen-Koskela 1998 ‘Quasiconformal maps in metric spaces of controlled geometry’; Albin-Brunner-Perez-Pisaroni-Poggi-Corradini 2015 ‘Modulus of families of walks on graphs’; Albin-Poggi-Corradini’s discrete-modulus school). The $p=3$ functional $\sum_e \rho(e)^3$ is convex but NOT a quadratic form, so it lies strictly between min-cut ($p=1$) and effective resistance ($p=2$), with a dual that is a Hölder-3 flow-cost program rather than a Sherman-Morrison pseudoinverse identity. Distinct from blacklisted effective-resistance / signed-Laplacian / Szegedy-walk / Ihara-zeta attempts: none of those is a $p \neq 2$ modulus. Uses sup/abs/cube-of-sums (no polynomial F_q extension, Aaronson-Wigderson-safe). An arXiv search ‘p-modulus’ AND (‘Tseitin’ OR ‘resolution refutation’ OR ‘DPLL’ OR ‘proof complexity’) returns 0 direct hits and <5 adjacent papers (all on Albin-Poggi-Corradini network modulus or QC analysis on fractals, never on CNF refutation). × Tree-like Resolution / DPLL refutation tree size $t(T(G, \omega))$ for Tseitin XOR formulas $T(G, \omega)$ on connected 3-regular graphs G with odd charge ω (Urquhart 1987 / Ben-Sasson-Wigderson 2001 / Alekhovich-Razborov 2001 expansion regime), in the Cook-Reckhow-Krajíček bounded-arithmetic framework where $\neg T(G, \omega)$ is Σ^b_0 and $\log_2 t$ lower-bounds V^0 -witnessing complexity. The invariant is STRUCTURAL on (G, ω) : two charged graphs whose Tseitin formulas are both $\equiv \text{FALSE}$ can have very different M_3 , shielding from Razborov-Rudich natural proofs.
- **Recorded:** 2026-05-18 10:26 UTC
- **Entry ID:** 67ed17c63684

Statement

Let G be a connected 3-regular graph on n vertices, $m=3n/2$ edges, and let $\omega: V(G) \rightarrow \mathbb{F}_2$ satisfy $\sum_v \omega(v)=1$ (odd-charge, so $T(G, \omega)$ is unsatisfiable). Let Γ_ω be the family of simple G -paths whose endpoints $u \neq v$ both have $\omega(u)=\omega(v)=1$. Define the discrete $p=3$ modulus $M_3(G, \omega) := \inf \{ \sum_e \rho(e)^3 : \rho: E(G) \rightarrow \mathbb{R}_{\geq 0} \text{ and } \sum_{e \in \gamma} \rho(e) \geq 1 \text{ for all } \gamma \in \Gamma_\omega \}$, computed in polynomial time by convex programming. Conjecture: there exists an absolute constant $c > 0$ such that for every such (G, ω) , $\log_2 t^*(T(G, \omega)) \geq c \cdot n \cdot M_3(G, \omega)$; furthermore for every (G, ω) with G a 3-regular expander (girth ≥ 4 , $\lambda_2(A_G) \geq 0.4$) and ω odd, $M_3(G, \omega) \geq 1/(c \cdot n^{2/3})$. A single (G, ω) violating either inequality with the calibrated $c=0.05$ refutes the conjecture.

Rationale

Tseitin hardness on expanders is governed by edge-disjoint flow-thickness between charged vertices, which the $p=2$ (effective resistance) modulus already captures only up to spectral information. The $p=3$ functional penalises concentrated densities CUBICALLY, so on expanders the optimal ρ is spread nearly uniformly, giving $M_3 \approx m \cdot m^{-3} \approx \Theta(n^{-2/3})$; on bottlenecked instances M_3 collapses toward a single-edge cost ≈ 1 . This non-quadratic response is invisible to all spectral / pseudoinverse / non-backtracking invariants already exhausted on Tseitin, and the cube-of-sums dual is precisely an $L^{3/2}$ -flow program — a regime never touched by the Ben-Sasson-Wigderson width / expansion machinery.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e01c7e79.py", line 267, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e01c7e79.py", line 243, in run_trial
    adj = sample_3_regular_graph(n, seed)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e01c7e79.py", line 134, in sample_3_reg
    if is_connected(adj) and girth(adj) >= 4 and spectral_gap(adj) >= 0.4:
                                   ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e01c7e79.py", line 112, in spectral_gap
    L = matrix_sub(D, adj)
        ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e01c7e79.py", line 35, in matrix_sub
    return [[A[i][j] - B[i][j] for j in range(len(A[0]))] for i in range(len(A))]
            ~~~~^^^
```

IndexError: list index out of range

Judge reasoning

The test crashed with an `IndexError` in `matrix_sub` during `spectral_gap` computation before any instances were evaluated, so no data on the two pre-registered metrics was produced. | next: Fix the matrix dimension bug in `sample_3_regular_graph/spectral_gap` (ensure `adj` is built as a full $n \times n$ matrix before Laplacian subtraction) and rerun the 120-instance protocol.